ONE-PAGE BRIEF

Post-quantum cryptography review: zingolib @ 81d799775f11

Client: Zingo Labs — public repository (passive review) · Source: github.com/zingolabs/zingolib, commit `81d799775f11` · Reviewed: 2026-11-04 19:45 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 3 of 4
hybrid protection in a critical path

FINDINGS
0 Critical **3** High **6** Medium
31 Low **11** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Cover the remaining classical critical roles and plan when the classical half can be dropped.

WHAT MATTERS MOST

High QSI-2026-13044	**pulldown-cmark 0.13.4 is affected by RUSTSEC-2026-0414** What to do: Upgrade to a fixed version (0.13.5).
High QSI-2026-13045	**pulldown-cmark 0.13.4 is affected by RUSTSEC-2026-0433** What to do: Upgrade to a fixed version (0.13.6).
High QSI-2026-13055	**rustls-webpki 0.101.7 is affected by GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104)** What to do: Upgrade to a fixed version (0.103.13, 0.104.0-alpha.7).
Medium QSI-2026-13047	**rustls 0.21.12 is affected by GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346)** What to do: Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

36 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium threshold signing and MPC protocols	Pedersen commitment	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the…
Medium transaction and message signing	ECDSA	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum…

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 40; unlikely to apply to this build: 17 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 23.

Static review of source code and manifests; files analysed: 205. No requests were sent to the owner's systems.

Full report, content SHA-256: `94ba853b5bcb28e94d33bd80aac4168e62a7586072ef696d16a4566906d061d6`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.